

LIFE ON EARTH

Earth's land surface, oceans, and atmosphere teem with life, which has colonized every corner of our planet. This is remarkable given that when Earth formed, it was just a dead rock. Life has drastically altered the appearance of our world and the chemistry of the atmosphere.

Early life on earth

Life on Earth probably originated around 4.25 billion years ago, from precursor chemicals. The oldest definite evidence of life is around 3.7 billion years old and suggests that simple single-celled organisms existed in oceans at that time. These microbes were already able to photosynthesize—that is, to use light energy to manufacture sugars from water and dissolved carbon dioxide, releasing oxygen in the process. By around 1.9 billion years ago, photosynthesizing organisms had released significant quantities of oxygen into the atmosphere. This was important to further evolution of life, because some of the oxygen (consisting of O_2 molecules) was converted into ozone (O_3 molecules), which formed a protective layer around the planet, filtering out harmful incoming ultraviolet rays. The presence of oxygen in the atmosphere, and dissolved in the oceans, also made animal life possible. By 1.2 billion years ago, multicellular organisms made from eukaryotic cells (more complex than the earlier prokaryotic cells of the simple microbes) had evolved.



◀ OLDEST SIGNS OF LIFE

The oldest fossils known are 3.7 billion-year-old traces of stromatolites—layered structures formed by microbes living on an ancient seafloor—within this rock found in Greenland.

Evolution, spread, extinctions

Between around 525 and 500 million years ago (MYA), during a time in Earth's history called the Cambrian Period, a great proliferation and diversification of life with mineralized shells and skeletons occurred in the oceans. Today, this is known as the Cambrian explosion. By 490 MYA, the first plants had appeared on land, and by 420 MYA, there were also animals on land. From then onward, life continued to spread and diversify rapidly, but there were also extinctions.

One major wave of extinctions occurred at the end of the Permian Period, around 252 MYA. The exact cause is not known, although a series of massive volcanic eruptions is one suspect. Another occurred at the end of the Cretaceous Period, 66 MYA, when more than 75 percent of all species died out, including most dinosaurs. However, one dinosaur group survived and evolved into birds.

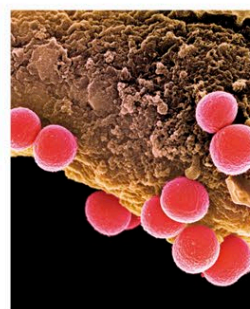
▷ EXTINCT FORM

Tylocidaris sea urchins, of which this is a fossil example, lived between about 140 million and 40 million years ago, but then died out.



Kingdoms of life today

Life on Earth is extremely diverse. Estimates of the number of Earth's current species range from 2 million to 1 trillion, of which only about 1.6 million have been identified and described so far. Various schemes have been put forward for dividing known life forms into kingdoms. A consensus scheme, proposed by scientists at the Smithsonian Institution in 2015, defines seven kingdoms. Two consist solely of



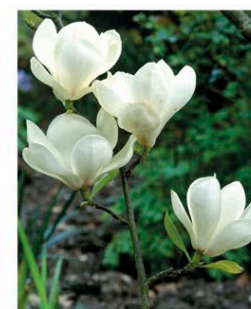
△ ARCHAEA

Many of these simple, single-celled lifeforms live in extreme environments, such as acidic hot springs.



△ PROTOZOA

Some protozoa cause disease. This one, *Giardia lamblia*, causes a type of human intestinal infection.



△ PLANTS

There are over 300,000 known species of plants. They are the basis of most ecosystems, especially on land.

Life has been found inside rocks 12 miles (19 km) below Earth's surface

prokaryote organisms—bacteria and archaea. The other five are based on eukaryotic cells: protozoa (single-celled life forms such as amoebae); chromista (algae of various sorts, including diatoms and water molds); and three more familiar groups: fungi, plants, and animals. The various life-forms interact with Earth, affecting

landscapes in many ways. For example, plants cover a significant proportion of the land surface, and their roots help keep soil from being carried away by rainwater. Life has also changed the chemistry of Earth's atmosphere, making it uniquely oxygen-rich. Without life, Earth would have evolved and would appear and work quite differently.

▷ RIVER CROSSING

One of the most awe-inspiring natural sights on Earth is the annual migration of vast numbers of wildebeest, zebras, and gazelles—here about to cross a river—across the Serengeti ecosystem.